Discrete Mathematics **Name (Print):** ____________________
MATH 208

- This exam contains 13 pages (including the front and back of this cover page) and 14 problems. Check to see if any pages are missing.
- This is the third exam and will count for 25% of your final grade. You have 120 minutes (1:00 pm - 3:00 pm) to complete this exam. You may use one index card of notes and a calculator on this exam, but no other outside sources may be consulted.

The following rules apply:

- Every problem on this exam, other than true-or-false problems or multiple choice problems, requires work of some form. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. **A correct answer that has no supporting work will not receive full credit.**
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- There are 140 points total in this exam, as well as 1 bonus questions worth a total of 5 possible points.

Page:	3	4	5	6	7	8	9	10	11	12	Total
Points:	20	10	15	24	8	10	10	15	10	18	140
Score:											

Bonus:	

1. Mark each statement as True or False by completely shading in the box corresponding to your answer.

2 pts (a) $242 \equiv -23 \mod 5$

True False

2 pts (b) $\binom{25}{12} = \binom{25}{13}$

True False

2 pts (c) When $(x-1)^{25}$ is expanded, the coefficient of x^{10} is $\binom{30}{10}$.

True False

2 pts (d) 759 days after a Monday is a Thursday.

True False

2 pts (e) In a group of 200 people, there are always at least 30 born on the same day of the week.

True False

2 pts (f) There are $10^5 - 10^4$ strings of length 10 over the alphabet $\Sigma = \{a, b, c, d, e\}$ that contain at least one a.

True False

2 pts (g) A graph with a Hamilton circuit has a Hamilton path.

True False

2 pts (h) A regular, connected graph that has 10 vertices and 40 edges has an Euler circuit.

True False

2 pts (i) The Nearest Neighbor Algorithm always gives the optimal tour of a graph with Hamilton circuits.

True False

2 pts (j) The Brute Force Algorithm always gives the optimal tour of a graph with Hamilton circuits.

True False

2. There are five roads from A to B, three roads from B to C, and five roads from C to D.

3 pts (a) How many different routes are there from A to B to C to D?

(A) $2(5) + 3$ **(B)** $(5)^2(3)$ (C) $(3^5)(3^3)(3^5)$ (D) $2(3^5) + 3^7$

3 pts (b) How many different round trip routes are there from A to D and back to A?

(A) $(5^2)(3^2)(5^2)$ (B) $2\big(2(5) + 3\big)$ (C) $(3^{10})(3^6)(3^{10})$ (D) $2\big(2(3^5) + 3^7\big)$

4 pts (c) How many different round trip routes are there from A to D and back to A if you cannot travel over any road more than once?

(A) $2(5) + 3 + 2(4) + 2$ (B) $(3^9)(3^5)(3^9)$ (C) $2(3^5) + 3^3 + 2(3^4) + 3^2$ **(D)** $(5^2)(3)(4^2)(2)$

5 pts 3. Computer ID's are length seven strings made up of any combination of seven different letters (a-z) and digits (0-9). How many different ID's are there?

(A) $(36)(7)$ (B) 36^7 (C) $\dfrac{36!}{(36-7)!}$ (**D**) $\dfrac{36!}{7!(36-7)!}$

4 pts 4. The number of strings of letters A-Z of length 10 that both begin with "A" and end with "Z" is:

(A) $2(26^8)$ (**B**) 26^8 (C) $26^9 + 26^5$ (D) $(26^9)(26^9)$

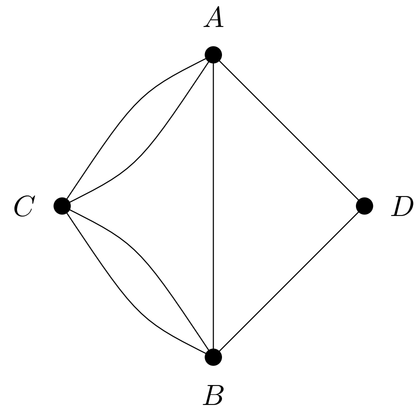
6 pts 5. Determine the coefficient of x^5y^2 in the expansion of $(2x - 5y)^7$.

(A) $\binom{7}{5}(-5)^5(2)^2$ (B) $\binom{5}{2}(2^5)(-5)^2$ (**C**) $\binom{7}{5}(2^5)(-5)^2$ (D) $\binom{7}{5}$

$\boxed{12 \text{ pts}}$ 6. For $n > 1$, in how many ways can a subset of $k > 1$ numbers from 1 to n (inclusive) be selected if the selected numbers cannot be consecutive?

$\boxed{12 \text{ pts}}$ 7. Suppose p, q are two different primes. How many integers between 1 and the product pq are relatively prime to pq (or, to rephrase things in a slightly different but equivalent way, how many integers between 1 and pq are divisible by neither of p and q)?

8. Use the graph, G, drawn below to answer the following questions.



1 pts (a) Write vertex set, $V(G)$.

1 pts (b) Write the edge set, $E(G)$.

1 pts (c) Which edges, if any, are incident to DB?

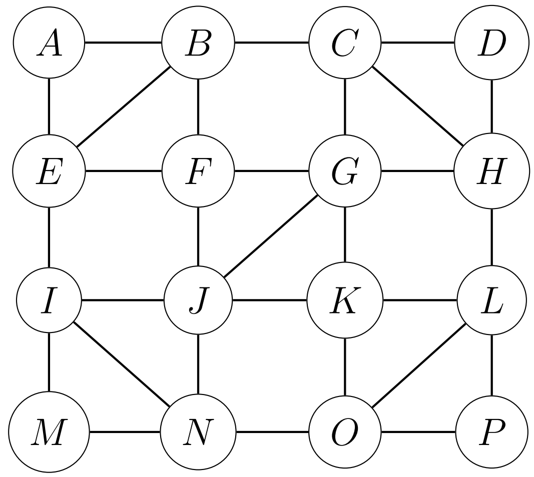
1 pts (d) Which vertices, if any, are adjacent to C?

1 pts (e) What is the degree of each vertex?

1 pts (f) Which edges, if any, are bridges?

2 pts (g) If an Euler Circuit or Euler Path exists, list one and specify whether it is an Euler circuit or an Euler path. If neither is possible, explain why.

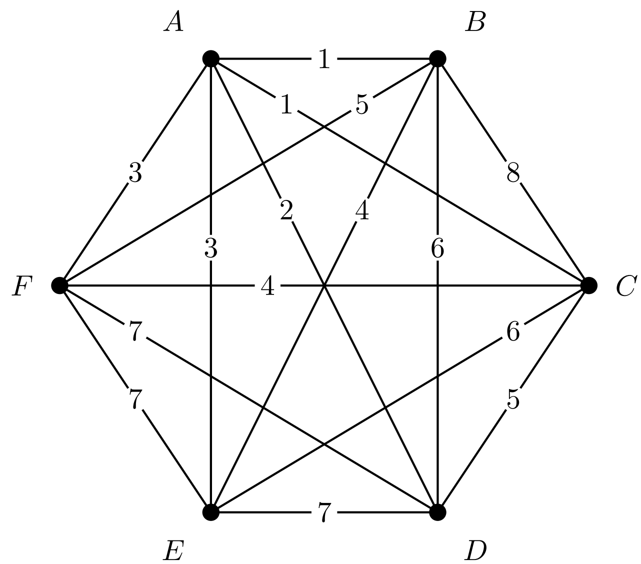
10 pts 9. Use Fleury's algorithm to find an Euler path in the graph below.



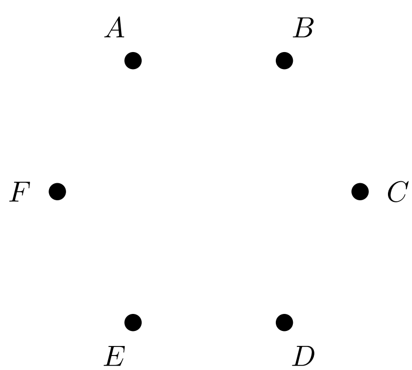
Show your answer by labeling the edges $1, 2, 3$ and so on in the order they are traveled.

10 pts 10. Occasionally, a situation may arise where the Nearest Neighbor Algorithm could produce multiple results. That is, it might be the case that there are multiple "cheapest" edges at a given vertex. The following weighted graph is an example of a situation where this might happen. Use the graph below to find two Hamilton circuits resulting from the Nearest Neighbor Algorithm (starting at F). For your convenience, the edge weights are also sorted.

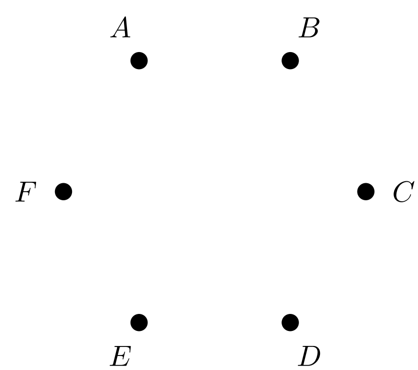
Edge	AB	AC	AD	AE	AF	BE	CF	BF	CD	BD	CE	DE	DF	EF	BC
Weight	1	1	2	3	3	4	4	5	5	6	6	7	7	7	8



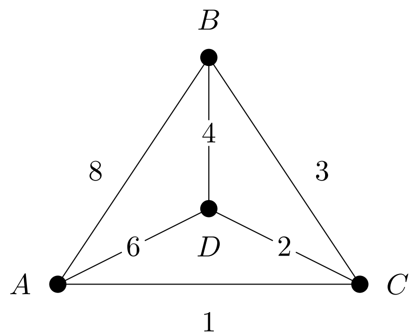
NNA tour 1 ✓ + −



NNA tour 2 ✓ + −

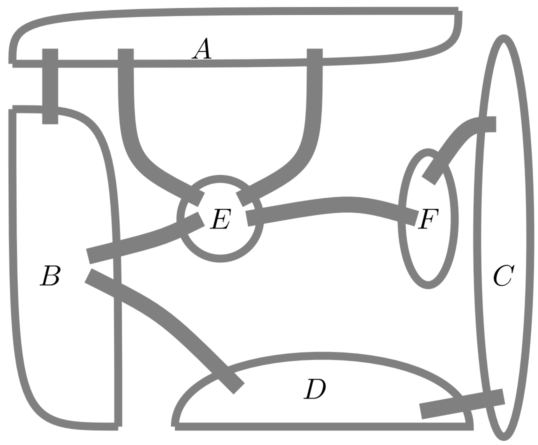


15 pts 11. Use the weighted graph below to demonstrate your prowess with the Repetitive Nearest Neighbor Algorithm.

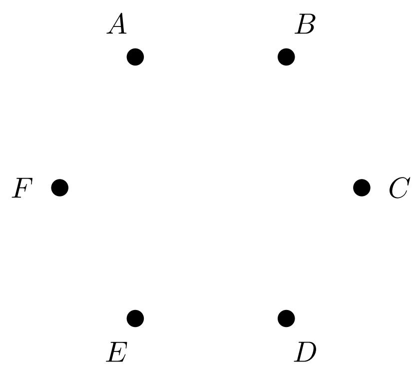


What is the route and weight resulting from the application of the Repetitive Nearest Neighbor Algorithm?

12. Use the map of islands and bridges shown below to answer the next three questions.



3 pts (a) Draw a graph that models the islands (as vertices) and bridges (as edges).



4 pts (b) Is it possible to start a journey at D and end at B, crossing each bridge in the picture exactly once? Explain with a mathematical reason. (Note: a list of vertices is *not* a mathematical reason.)

3 pts (c) If your answer in **(b)** was 'Yes', list the path. If your answer was 'No', explain whether or not is it possible to start a journey at some other location and end at B while crossing each bridge in the picture exactly once.

13. In this problem, you are presented with a number of small drawing questions. You do not need to label the vertices. If you are directed to draw a graph that is impossible, say so and explain why. If the number of vertices is not specified, use no more than six; if the number of edges is not specified, use no more than eight.

3 pts (a) Draw a graph that has a Hamilton Path but no Euler Path.

3 pts (b) Draw a graph that has exactly four vertices and exactly two bridges.

3 pts (c) Draw a graph that is not simple and not regular.

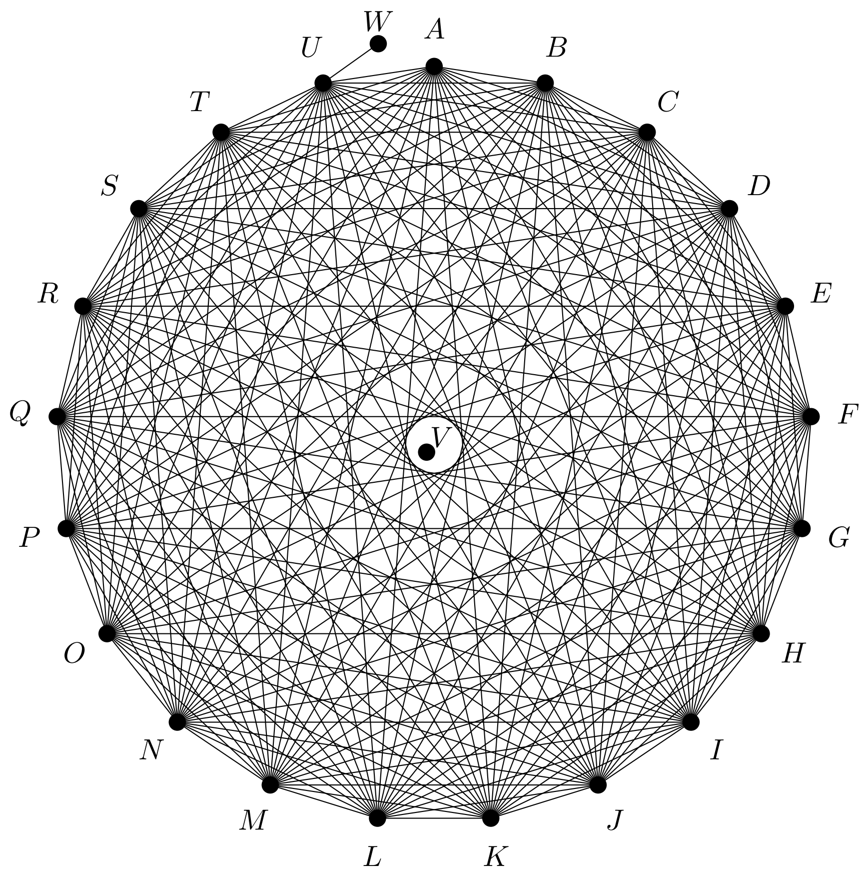
3 pts (d) Draw a graph that is regular and connected, but has no Euler Paths, Euler Circuits, Hamilton Paths, or Hamilton Circuits.

3 pts (e) Draw an example of a weighted graph with four vertices where the Brute Force Algorithm is quicker (fewer Hamilton Circuits are checked) than the Repetitive Nearest Neighbor Algorithm. (Hint: there is a *very easy* example of this)

3 pts (f) Draw an example of a regular graph that has at least three vertices and an Euler Path.

5 (bonus) 14. Use the graph below to answer the next five questions. If you aren't sure how many of something there are, at least give a range of possible values with an explanation.

(Note: It might be easy to miss, so be sure to notice the vertex V in the center of the graph)



(a) How many bridges are in this graph?

(b) How many Euler Paths are in this graph?

(c) How many Euler Circuits are in this graph?

(d) How many Hamilton Paths are in this graph?

(e) How many Hamilton Circuits are in this graph?